NEXTUTE EDTECH PVT. LTD. × FINANCE CLUB, IIT PATNA

Case Competition 2026 · Participant Guide

CASE B

The Data Moat

Build an AI-powered intelligence platform that turns the messy data of India's coaching institutes into better results — and a profitable SaaS business

> *"Thousands of institutes run on WhatsApp, Excel and OMR sheets. The data is there. The intelligence is not."*

FORMAT

Round 1 Online → Round 2 Offline

WEIGHTAGE

70% Technology · 30% Edtech market

ELIGIBILITY

Students of IITs & IIMs

Read this document fully before starting. The same problem statement applies to both rounds.

SECTION 01

About the Competition

This case competition is jointly organised by **Nextute Edtech Private Limited** and the **Finance Club, IIT Patna**. It is designed for students of IITs and IIMs who want to solve a real, open-ended problem at the intersection of **technology, finance and education**.

You are reading the guide for **CASE B: The Data Moat**. A separate guide exists for Case A: Learn Now, Pay After Placement (AI underwriting for outcome-based education). Each team works on only one case.

Item	Details
Organisers	Nextute Edtech Pvt. Ltd. × Finance Club, IIT Patna
Who can participate	Currently enrolled students of any IIT or IIM (UG, PG, PhD, online/hybrid programmes included)
Team size	2 to 4 members. Cross-institute and cross-discipline teams are allowed and encouraged.
Rounds	Round 1: Online submission · Round 2: Offline finale (shortlisted teams)
Problem statement	**The same problem statement applies to both rounds.** Round 2 goes deeper: a working prototype, a detailed model and a live surprise scenario.
Weightage	**70% technology** (system design, data, ML/AI, engineering) · **30% edtech market** (business model, unit economics, go-to-market)
Rewards	To be announced by the organisers

How to read this guide

- Sections 2–4 explain the context and the exact problem you must solve.
- Sections 5–6 describe what a complete solution should contain and the revenue models you can explore.
- Section 7 gives an illustrative data pack and assumptions so every team starts from the same baseline.
- Sections 8–11 cover resources, deliverables for each round, judging criteria and rules.

SECTION 02

Background & Context

Test-prep coaching for exams like JEE and NEET is one of the largest segments of Indian education. While a few national brands run sophisticated tech platforms, a very large part of the market consists of **small and mid-sized local institutes** in cities like Patna, Kota, Lucknow, Jaipur, Indore and hundreds of smaller towns.

These institutes generate a lot of data — weekly tests, attendance, fee payments, doubts, homework — but most of it lives in **paper OMR sheets, Excel files, registers and WhatsApp groups**. As a result:

- Teachers do not know which topics each student is weak in until it is too late.
- Students who are about to drop out or stop paying fees are noticed only after they leave.

- Parents get little visibility beyond a marks list, and trust suffers.
- Owners cannot measure which batches, faculty or teaching methods actually work.
- Local institutes struggle to compete with the data-driven personalisation of large national players.

Why this case, why now

Nextute works with coaching institutes and provides them a dashboard app to automate lecture scheduling and tests. The next step is clear: **move from operations software to intelligence** — products that predict, personalise and prove results. Whoever collects clean, structured learning data across many institutes builds a **data moat**: better models, better results, more institutes, more data.

> **The core tension**
>
> Institute owners are price-sensitive, staff are often not tech-savvy, internet can be patchy, and most students are minors whose data needs special protection. Your platform must be powerful *and* simple, cheap, trustworthy and compliant.

SECTION 03

The Problem Statement

> **Your mission**
>
> *Design and build an **AI-powered Institute Intelligence Platform** for small and mid-sized coaching institutes in India. The platform must (a) capture data from the messy, real-world sources institutes use today, (b) turn it into predictive and personalised insights — such as topic-level weakness, dropout risk and expected exam performance — for owners, teachers, students and parents, and (c) be packaged as a SaaS business that institutes will pay for and that can scale profitably across India.*

Key questions you must answer

A. Technology (70%)

1. **Data capture:** How will you get data from paper OMR sheets, Excel, Google Forms and WhatsApp into your system with minimum effort from staff? How does it work on low bandwidth or offline?
2. **Data model:** How will you structure students, batches, tests, questions, syllabus topics, attendance, fees and doubts so that insights are possible? How will questions be tagged to topics at scale?
3. **Learning analytics:** How will you estimate each student's mastery of each topic (e.g. knowledge tracing or item response theory) and calibrate question difficulty?
4. **Early-warning system:** How will you predict which students are likely to drop out, fall behind or stop paying fees — early enough to act?
5. **Performance prediction:** How will you estimate a student's likely exam rank/percentile range, and how will you communicate its uncertainty honestly?
6. **Generative AI layer:** How can LLMs generate personalised study plans, parent reports in Hindi/English, practice questions or doubt answers — reliably and at low cost?

7 **Architecture:** Show a multi-tenant, secure, low-cost architecture. What is your infrastructure + AI cost per student per month?

8 **Privacy & safety:** Most users are minors. How do you handle consent, data ownership, access control and student well-being?

B. Edtech market & business (30%)

1 **Customer & value:** Who is the buyer (owner, director, parent?) and what is the measurable ROI for an institute?

2 **Pricing & unit economics:** What do you charge, how is it packaged, and what are CAC, payback, churn and LTV?

3 **Go-to-market:** Which cities and institute sizes first? Direct sales, channel partners or product-led growth?

4 **Competition & moat:** How do you win against generic institute-management apps and ERPs, and why does your advantage grow over time?

SECTION 04

Scope & Weightage (70 : 30)

Your solution will be judged roughly **70% on technology** and **30% on the edtech market and business**.

Component	Weight	What it covers
System architecture & engineering	25%	Data ingestion, data model, multi-tenancy, APIs, offline-first design, security, cost per student
Data science, ML & AI rigour	25%	Knowledge tracing / IRT, dropout prediction, rank estimation, GenAI quality controls, validation
Feasibility, privacy & safety	20%	Real-world usability for low-tech staff, DPDP compliance for minors, well-being safeguards, reliability
Revenue model & unit economics	20%	Pricing, packaging, CAC, payback, churn, LTV, 3-year projection, sensitivity
Market & go-to-market	10%	Segment choice, ROI story for institutes, sales channel, competition, moat

Out of scope

- Building full ERP features (fee collection, payroll, timetabling) from scratch — assume basic operations data exists or can be imported.
- Creating complete study content or question banks.
- A production mobile app. A working web prototype is enough.

SECTION 05

What a Complete Solution Looks Like

There is no single correct answer. The flow below is a **reference skeleton** showing the expected depth — strong teams will challenge and improve it.



Expected solution components

Layer	What we expect to see	Ideas to explore
Data ingestion	Realistic capture methods that need minimal staff effort	Phone-camera OMR scanning (OpenCV), Excel/CSV templates, Google Forms sync, WhatsApp Business bot, offline queue + sync
Data model	A clear schema linking students, questions, topics and outcomes	Syllabus topic graph (JEE/NEET), question–topic tagging with an LLM + human review, multi-tenant PostgreSQL
Learning analytics	Topic-level mastery per student; question difficulty	Item Response Theory (IRT), Bayesian Knowledge Tracing (BKT), Deep Knowledge Tracing (DKT)
Early warning	Risk scores with reasons, early enough to act	Gradient boosting or survival models on attendance, score trends, fee delays, engagement drops
Performance prediction	Rank / percentile ranges with uncertainty	Calibrating internal test scores against mock / previous-year distributions; prediction intervals
GenAI layer	Useful, accurate, cheap AI features with guardrails	RAG over syllabus + institute content, bilingual parent reports, auto-generated practice sets, answer verification
Dashboards & actions	Different views for owner, teacher, student, parent	Topic heatmaps, at-risk lists, batch comparisons, WhatsApp-delivered weekly reports
Business model	Pricing, packaging, unit economics and rollout plan	Per-student SaaS, tiered plans, AI add-ons, parent subscriptions, outcome-linked pricing

Student well-being is a design requirement

Coaching students are under heavy pressure. “At-risk” flags must lead to **supportive** actions (mentoring, counselling, remedial help) — never to public ranking, shaming or pressure on the student. Show how your design protects students.

SECTION 06

Revenue Models to Explore

You may choose one model or combine several. Justify your choice using unit economics and the institute's willingness to pay.

Model	How it works	Strengths	Risks / challenges
Per-student SaaS	Institute pays ₹X per active student per month	Scales with institute size; simple	Price-sensitive buyers; seasonal student counts
Tiered plans	Basic (operations) / Pro (analytics) / Premium (AI)	Easy upsell path; captures different budgets	Feature gating must feel fair
Freemium + AI add-ons	Core free; pay for AI reports, rank prediction, study plans	Fast adoption; builds the data moat	Low conversion; free users cost money
Outcome-linked pricing	Fee tied to retention improvement or results	Strong ROI story; aligned incentives	Hard to measure attribution; disputes
Parent subscription (B2B2C)	Parents pay for premium progress reports and insights	Second revenue stream; high perceived value	Institute may resist; collection effort
Marketplace / upsell	Test series, content, mentorship sessions, lead generation for institutes	High-margin add-ons on the same platform	Distracts from core; quality control
Aggregated benchmarks	Anonymised city/state-level performance benchmarks for institutes	Unique value from the data moat	**Strict privacy limits**; must never expose individual data

A naïve worked example — and why it is not enough

Institute side (ROI): an institute has 400 students paying ₹60,000/year over 10 months (₹6,000/month). If 12% drop out around month 4, each lost student costs ₹36,000 of remaining fees.

Step	Calculation	Value / year
Fees lost to dropouts	400 × 12% × ₹36,000	₹17,28,000
Saved if platform prevents 25% of dropouts	12 students × ₹36,000	₹4,32,000
Platform cost at ₹40/student/month	400 × ₹40 × 12	₹1,92,000
Return on platform cost	₹4,32,000 ÷ ₹1,92,000	**≈ 2.25×**

Platform side (unit economics):

Step	Calculation	Value
Monthly revenue per institute	400 × ₹40	₹16,000
Infra + AI cost	400 × ₹6	₹2,400
Support & customer success	Allocated per institute	₹3,000
Contribution margin / month	₹16,000 – ₹5,400	₹10,600

Step	Calculation	Value
CAC + onboarding	₹25,000 + ₹8,000	₹33,000 → payback ≈ 3.1 months
LTV at 3% monthly churn	₹10,600 × ~33 months	**≈ ₹3.5 lakh (LTV/CAC ≈ 10×)**

Your job is to challenge this

These numbers look very attractive — which is exactly why judges will question them. Will owners really pay ₹40 per student? Will the platform actually prevent 25% of dropouts, and how would you prove it? What happens to churn after the exam season? What does it cost to onboard a low-tech institute? Build a model with ranges, scenarios and evidence.

SECTION 07

Illustrative Data Pack & Assumptions

All values are **illustrative baselines**. You may change them with justification.

Typical institute profile

Parameter	Baseline value	Notes
Students per institute	400 (range 150–2,000)	Split into 5–20 batches
Annual fee per student	₹60,000 (range ₹30,000–1,50,000)	Usually paid in instalments
Faculty & staff	12–25 teachers, 2–5 admin staff	Low to moderate tech comfort
Tests	2–4 tests per batch per month	Mostly paper-based OMR + some online
Current tools	Excel, registers, WhatsApp groups, basic apps	Data scattered and inconsistent
Mid-session dropout / fee default	8–15%	Highest in first 4–5 months
Languages	Hindi and English (regional in some states)	Parent communication often in Hindi
Connectivity	Mobile-first; patchy internet in smaller towns	Offline-first design helps

SaaS business assumptions

Parameter	Baseline value
Price range to explore	₹20–80 per student per month (or equivalent plan pricing)
CAC per institute	₹15,000–30,000 (field sales + demos)
Sales cycle	3–8 weeks; peaks before new session (March–June)
Onboarding cost per institute	₹5,000–10,000 (data migration, training)
Infrastructure + AI cost	₹3–10 per student per month (depends on LLM usage)
Monthly logo churn	2–5%

Suggested data schema (for synthetic data)

You will not receive real institute data. Build a synthetic dataset (recommended: 3+ institutes, 1,000+ students, a full academic year of tests) using this schema and/or the public datasets in Section 8. Explain your assumptions.

Entity	Example fields
institutes	institute_id, city, city_tier, exam_focus (JEE/NEET), student_count, plan
students	student_id, institute_id, batch_id, class (11/12/dropper), join_date, fee_plan, parent_contact_consent
questions	question_id, subject, chapter, topic, sub_topic, difficulty (estimated), source
test_responses	test_id, student_id, question_id, response, is_correct, time_taken_sec, test_date
attendance	student_id, date, session_id, present_flag
fees	student_id, instalment_no, due_date, paid_date, amount
engagement	student_id, date, doubts_asked, homework_submitted, app_logins
outcomes (labels)	dropped_out_flag, dropout_month, final_mock_percentile, actual_exam_percentile

SECTION 08

Resources & Learning Material

These are starting points, not requirements. Always cite what you use.

Public datasets

Dataset	Where	Useful for
EdNet (Riiid)	GitHub / research release	Large-scale question-level responses; knowledge tracing
ASSISTments datasets	ASSISTments research data	Knowledge tracing benchmarks
Open University Learning Analytics Dataset (OULAD)	Open University (UK)	Dropout / at-risk prediction from engagement data
Student Performance	UCI Machine Learning Repository	Performance prediction baseline
xAPI-Edu-Data (Students' Academic Performance)	Kaggle	Engagement features and outcomes
JEE/NEET syllabus & past papers	NTA official website	Topic graph and question tagging

Tools & libraries

Purpose	Suggested tools
Data capture	OpenCV (OMR scanning), Tesseract OCR, WhatsApp Business Platform (Cloud API), Google Forms / Sheets API
Learning analytics	pyBKT, pyKT (knowledge tracing), girth / py-irt (Item Response Theory)
Prediction	scikit-learn, XGBoost / LightGBM, lifelines (survival analysis)

Purpose	Suggested tools
GenAI	LLM APIs or open models, LangChain / LlamaIndex for RAG, evaluation with test sets
Data & dashboards	PostgreSQL, dbt, Metabase / Apache Superset, Streamlit
Prototype	Streamlit, Gradio, React / Next.js, FastAPI, Docker

Concepts worth reading about

- Item Response Theory (1PL / 2PL / 3PL models) and computerised adaptive testing.
- Knowledge tracing: BKT, DKT, and attention-based models (e.g. SAKT).
- Early-warning systems in education and dropout prediction.
- Multi-tenant SaaS architecture and cost-per-tenant modelling.
- SaaS metrics: CAC, payback period, gross margin, net revenue retention, LTV/CAC.
- Network effects and data moats — and their limits.

Regulatory & ethical areas to address

Participants must check the latest official versions — rules change. Show how each affects your design.

Area	Why it matters to your design
Digital Personal Data Protection Act, 2023 & DPDP Rules, 2025	Most students are under 18 → verifiable parental consent; the Act restricts tracking and behavioural monitoring of children. You must explain how features like dropout prediction stay within the law.
Ministry of Education guidelines for coaching centres (2024)	Registration, student well-being, no misleading claims about results
Data ownership	Who owns the data — the institute, the student or the platform? Can it be used across institutes?
Ethics & well-being	Supportive interventions only; no shaming; careful communication of predictions to students and parents

SECTION 09

Deliverables for Each Round

Both rounds use the same problem statement. Round 1 tests **whether your idea is sound**. Round 2 tests **whether it actually works** and whether you can defend it under pressure.

Round 1 · Online

#	Deliverable	Format & limits
1	Solution deck covering problem framing, solution architecture, data/ML approach, revenue model, unit economics and go-to-market	PDF, max 12 content slides (+ title & appendix). 16:9.
2	Financial model with clearly listed assumptions and at least one sensitivity analysis	Excel or Google Sheets (view link). Formulas must be live, not hard-coded.
3	Optional: notebook or code link showing an early model (e.g. knowledge tracing or dropout prediction) on public / synthetic data	GitHub / Colab link (bonus, not mandatory)

Round 2 · Offline finale

#	Deliverable	Format & limits
1	**Working prototype**: upload test results (CSV or an OMR sheet photo) → the system shows a topic-mastery heatmap, an at-risk student list with reasons, a predicted performance range, and an auto-generated parent report in Hindi or English.	Live demo during pitch; code repository link submitted in advance
2	Updated deck incorporating judges' Round 1 feedback	PDF, max 15 content slides
3	Detailed financial model (3-year projection, scenarios, sensitivities)	Excel / Google Sheets
4	Model & system report: validation metrics, cost per student, privacy design, known limitations	2–4 page PDF or appendix
	Curveball scenario: on the day, each team receives a new scenario (e.g. a regulatory, market or technical change). You must adapt your solution and present your response.	Approx. 60 minutes to prepare; presented within the pitch
	Pitch + Q&A	10 minutes pitch + demo, 5–7 minutes Q&A

What "working prototype" means

It does not need to be production-grade. It must run end-to-end on real or synthetic data and show the core logic live — not just screenshots or a Figma mock-up. A simple Streamlit / Gradio / web app backed by a real model is enough. Judges may ask you to change an input live.

SECTION 10

Judging Criteria

The same rubric is used in both rounds. In Round 1, judges evaluate the **soundness of the design**; in Round 2, they evaluate the **working prototype, depth of analysis and handling of the curveball**.

Area	Criterion	What judges look for	Weight
Technology (70%)	Architecture & engineering	Practical data capture, sound data model, multi-tenancy, offline-first, security, cost per student; in Round 2, quality of the working prototype	25%
	Data science, ML & AI rigour	Correct use of IRT / knowledge tracing, validated risk models, honest uncertainty, reliable GenAI with guardrails	25%
	Feasibility, privacy & safety	Usable by low-tech staff, DPDP-compliant for minors, well-being safeguards	20%
Market (30%)	Revenue model & unit economics	Pricing logic, institute ROI, CAC, payback, churn, LTV, 3-year projection, sensitivity	20%
	Market & go-to-market	Segment choice, sales channel, competition, defensibility of the moat	10%

Presentation quality and clarity are assessed within each criterion. Teams that clearly state their assumptions and limitations score higher than teams that oversell.

SECTION 11

Rules, Timeline & Submission

Timeline

Milestone	Date	Mode
Registrations open	To be announced	Online
Round 1 submission deadline	To be announced	Online upload
Round 1 results / shortlist	To be announced	Email + social handles
Round 2 finale	To be announced	Offline (venue to be announced)

Rules

1. All work must be original and created by the team for this competition. Plagiarism or copied solutions lead to disqualification.
2. AI tools (ChatGPT, Claude, Copilot, etc.) may be used, but you must disclose how you used them on the final slide. You must be able to explain every part of your solution.
3. Public datasets, open-source libraries and research may be used with proper citation. Do not use any personal data of real people without consent.
4. All numbers in this guide are **illustrative**. You may change any assumption if you justify it with evidence.
5. A member may be part of only one team. Team composition cannot change after the Round 1 deadline.
6. Late submissions will not be evaluated. File names must follow: **TeamName_Case_Round.pdf**.
7. By submitting, teams grant Nextute Edtech Pvt. Ltd. and the Finance Club, IIT Patna a non-exclusive right to use the submitted ideas and materials, with credit to the team where publicly showcased. Teams keep ownership of their code.
8. The judges' decision is final. The organisers may update rules; all updates will be communicated to registered teams by email.

Contact

Nextute Edtech Pvt. Ltd.: **admin@nextute.com** · Finance Club, IIT Patna: contact details will be shared with registered teams.

Frequently Asked Questions

Question	Answer
Will we get real institute data?	No. Use public datasets and synthetic data. Clearly explain how you generated it.
Do we need to build OMR scanning?	Not mandatory, but realistic data capture is a big part of this case. A simple working version earns credit.
Do we have to use GenAI?	Not mandatory. Use it where it adds clear value, and show how you control quality and cost.
Can we target institutes other than JEE/NEET?	Yes (e.g. UPSC, SSC, banking, school tuition), if you justify the market choice.
Will the Round 2 problem be different?	No. The problem is the same; Round 2 needs a working prototype, deeper analysis and a response to a surprise scenario.